

Active RIS-Assisted Integrated Sensing and Communication Systems: Joint Receive-Transmit Beamforming and Reflection Design

Yuantong Zhang¹, Junzhe Lin¹, Yifeng Zhao^{1,*}, Lianfen Huang¹, Zhibin Gao², Guozhen Xu¹

¹Department of Informatics and Communication Engineering, Xiamen University, Fujian, China

²Navigation Institute, Jimei University, Xiamen, China

Email: {zhangyuantong, linjunzhe, xuguozhen}@stu.xmu.edu.cn, {zhaoyf, lfhuang}@xmu.edu.cn, gaozhibin@jmu.edu.cn

Abstract—Active reconfigurable intelligent surface (RIS) acts as an enhancement of signal transmitted by the base station (BS) to overcome the effects of multiplicative fading and provide communication services for users. In this paper, we investigate an active RIS-assisted integrated sensing and communication (ISAC) system in which the BS senses the target in the presence of interference and establishes communication with multiple users assisted by the RIS. The optimization problem is formulated to jointly design the BS's sensing transceiver beamforming, communication beamforming, and the RIS's reflection coefficients to maximize the sensing SINR while satisfying the communication SINR requirements of users. An algorithm based on the ideas of quadratic transformation (QT), Karush-Kuhn-Tucker (KKT) conditions, semidefinite relaxation (SDR), and alternating optimization (AO) is designed to transform and solve the non-convex problem optimally. Simulation results verify the effectiveness and superiority of active RIS-assisted ISAC systems in improving sensing performance with communication SINR constraints.

Index Terms—Integrated sensing and communication (ISAC), reconfigurable intelligent surface (RIS), transmit and receive beamforming design.

I. INTRODUCTION

Integrated sensing and communication (ISAC) and reconfigurable intelligent surface (RIS) are two promising technologies with significant implications for future autonomous driving and the sixth generation (6G) wireless communication systems. ISAC technology allows communication and radar systems to share spectrum resources, offering high spectral efficiency and low costs, making it a potential technology for Vehicle-to-Everything (V2X) [1]. RIS is composed of many passive reflection elements that can dynamically adjust the phase of incident signals, establishing a reliable virtual link between the base station (BS) and users, making it a significant research hotspot.

Due to the advantages of RIS, there have been many research studies exploring the deployment of active RIS in ISAC systems [2]–[4]. The work [2] used RIS to assist multi-user communication and target sensing, maximizing the sum rate of communication users and meeting the worst-case

sensing signal-to-noise ratio (SNR). The work [3] aimed to maximize the weighted summation of the target sensing SNR under each user's communication quality of service (QoS) requirement. The work [4] has taken into account the presence of interference sources around the target. It maximizes the sensing signal-to-interference-and-noise-ratio (SINR) while minimizing multi-user communication interference through a joint waveform and transmit beamforming design. However, although the above work considered the existence of sensing interference in realistic scenarios, it still faces the challenge of multiplicative fading associated with passive RIS used for communication assistance [5]. Active RIS equipped with active loads on reflection elements has been proposed to reflect and amplify signals, thereby enhancing system performance [6]. Therefore, deploying active RIS in ISAC systems with sensing interference is a promising strategy. In addition, in order to better meet the needs of future intelligent traffic scenarios [7], we further study the joint design of sensing transmit and receive beamforming in the RIS-assisted ISAC system [8].

Motivated by the discussions mentioned above, in this paper, we investigate an active RIS-assisted ISAC system. The ISAC BS senses the target of interest (such as vehicles, unmanned aerial vehicles, and pedestrians) with nearby sources of interference. Simultaneously, due to obstructed line-of-sight (LoS) links between the BS and users, the BS provides communication services for multiple users with the assistance of active RIS.

The remainder of this article unfolds as follows: Section II presents the system model and the formulated problem. In Section III, an algorithm for transforming and solving the problem is proposed. Section IV presents the numerical results. Finally, Section V concludes this paper.

II. SYSTEM MODEL

We consider an active RIS-assisted ISAC system with sensing interference sources shown in Fig. 1. The ISAC BS is equipped with a uniform linear array (ULA) comprising N_t transmit antennas and N_r receive antennas aimed at detecting a target of interest while also providing communication services

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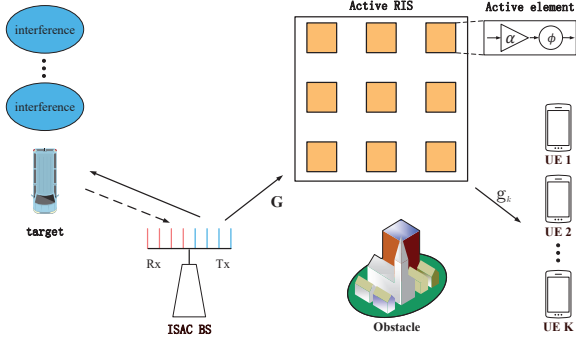


Fig. 1. An active RIS-assisted ISAC system.

to K single-antenna users. Without loss of generality, we assume the same number of transmit and receive antennas, i.e., $N_t = N_r = N$. There are L interference sources around the target of interest. By jointly designing the transmit and receive beamforming, the aim is to maximize the sensing SINR at the BS. Additionally, an RIS equipped with M reflecting elements assists the downlink communication, ensuring that users achieve the required communication quality even when obstructed by LoS obstacles.

The signal transmitted at the ISAC BS is expressed as

$$\mathbf{x} = \sum_{k=1}^K \mathbf{f}_k s_k + \mathbf{f}_r s_r, \quad (1)$$

where $s_k \sim \mathcal{CN}(0, 1)$ denotes the communication symbol intended to the k -th user, and s_r denotes the radar signal with zero mean and unit variance (normalized power) transmitted by the ISAC BS. $\mathbf{f}_k \in \mathbb{C}^{N \times 1}$ and $\mathbf{f}_r \in \mathbb{C}^{N \times 1}$ denote the corresponding beamforming matrices for communication symbols and radar signals, respectively.

A. Communication Model

Assume that multiple reflections are ignored due to severe channel fading. The signal received by the k -th user is expressed as

$$y_k = \mathbf{g}_k^H \Phi \mathbf{G} \mathbf{x} + \mathbf{g}_k^H \Phi \mathbf{z}_0 + n_k, \quad k = 1, \dots, K, \quad (2)$$

where $\mathbf{G} \in \mathbb{C}^{M \times N}$ and $\mathbf{g}_k \in \mathbb{C}^{M \times 1}$ represent the channels for the BS-RIS and RIS-UE links, respectively. $\Phi = \text{diag}(\mathbf{v})$ denotes the reflection coefficient matrix of the active RIS, where $\mathbf{v} = [\alpha_1 e^{j\phi_1}, \dots, \alpha_M e^{j\phi_M}]^T$, with $|\mathbf{v}_m| \leq \alpha_{\max}, \forall m$. $\mathbf{z}_0 \sim \mathcal{CN}(\mathbf{0}_M, \sigma_a^2 \mathbf{I}_M)$ represents the thermal noise at the active RIS [5], and $n_k \sim \mathcal{CN}(0, \sigma_u^2)$ denotes the additive white Gaussian noise (AWGN) at the k -th user.

Accordingly, the SINR for the k -th user is written as

$$\text{SINR}_{c,k} = \frac{|\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_k|^2}{\sum_{j \neq k} |\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_j|^2 + |\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_r|^2 + \sigma_a^2 \|\mathbf{g}_k^H \Phi\|^2 + \sigma_u^2}, \quad (3)$$

where $|\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_r|^2$ represents the impact of sensing on communication.

B. Radar Model

The ISAC BS senses a target of interest amidst interference. It is worth noting that the location information of both the target and the interference is known a priori [4], [9], [10]. The received echo signal at the BS is represented as

$$\mathbf{w}^H \mathbf{y} = \eta_0 \mathbf{w}^H \mathbf{A}(\theta_0) \mathbf{f}_r + \sum_{i=1}^L \eta_i \mathbf{w}^H \mathbf{A}(\theta_i) \mathbf{f}_r + \mathbf{w}^H \mathbf{n}_r, \quad (4)$$

where $\mathbf{w} \in \mathbb{C}^{N \times 1}$ denotes the beamforming at the BS, satisfying $\|\mathbf{w}\|^2 = 1$. The matrix $\mathbf{A}(\theta) = \mathbf{a}(\theta) \mathbf{a}^H(\theta)$, with the radar array steering vector $\mathbf{a}(\theta) = [1, e^{j \frac{2\pi d}{\lambda} \sin \theta}, \dots, e^{j \frac{2\pi (M-1)d}{\lambda} \sin \theta}]^T$, where $d = \frac{\lambda}{2}$. η_0 and η_i ($i \in \{1, \dots, L\}$) represent the complex amplitude coefficients of the target and the i -th interfering source, respectively. $\mathbf{n}_r \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ represents the AWGN at the BS. Therefore, the sensing SINR is given by

$$\text{SINR}_s = \frac{|\eta_0 \mathbf{w}^H \mathbf{A}(\theta_0) \mathbf{f}_r|^2}{\sum_{i=1}^L |\eta_i \mathbf{w}^H \mathbf{A}(\theta_i) \mathbf{f}_r|^2 + \sigma^2}. \quad (5)$$

C. Problem Formulation

We study an ISAC system focusing on the sensing function. The RIS assists multi-user communication when the LoS link is blocked, ensuring a certain level of communication quality. Our goal is to maximize the sensing SINR. It is achieved by jointly optimizing the communication transmit beamforming $\mathbf{f}_k$, the sensing transmit and receive beamforming $\mathbf{f}_r$ and $\mathbf{w}$, as well as the reflection coefficient matrix Φ . Therefore, the problem is formulated as

$$\begin{aligned} \text{(P0)} : \quad & \max_{\mathbf{w}, \mathbf{f}_k, \mathbf{f}_r, \Phi} \text{SINR}_s \\ \text{s.t.} \quad & \|\mathbf{w}\|^2 = 1 \end{aligned} \quad (6a)$$

$$\sum_{k=1}^K \|\mathbf{f}_k\|^2 + \|\mathbf{f}_r\|^2 \leq P_{\text{BS}} \quad (6b)$$

$$\sum_{k=1}^K \|\Phi \mathbf{G} \mathbf{f}_k\|^2 + \|\Phi \mathbf{G} \mathbf{f}_r\|^2 + \sigma_a^2 \|\Phi\|_F^2 \leq P_{\text{ARIS}} \quad (6c)$$

$$\text{SINR}_{c,k} \geq \gamma_{th,k}, \quad \forall k \quad (6d)$$

$$|\Phi_{m,m}| \leq \alpha_{\max}, \quad m = 1, \dots, M, \quad (6e)$$

where P_{BS} and P_{ARIS} represent the transmit power of the BS and the maximum power of the RIS, respectively. $\gamma_{th,k}$ denotes the communication SINR threshold for the k -th user.

III. PROPOSED SOLUTIONS

To address the high coupling and non-convexity issues in (P0), we first employ the quadratic transformation (QT) technique to process the objective function. By introducing the auxiliary variable a , (P0) can be recast as

$$\begin{aligned} \text{(P1)} : \quad & \max_{\mathbf{w}, \mathbf{f}_k, \mathbf{f}_r, \Phi, a} D(\mathbf{w}, \mathbf{f}_k, \mathbf{f}_r, \Phi, a), \\ \text{s.t.} \quad & \text{(6a)-(6e)}, \end{aligned} \quad (7)$$

where

$$D(\mathbf{w}, \mathbf{f}_k, \mathbf{f}_r, \Phi, a) = 2\Re \{ a^* \eta_0 \mathbf{w}^H \mathbf{A}(\theta_0) \mathbf{f}_r \} - |a|^2 \left(\sum_{i=1}^L |\eta_i \mathbf{w}^H \mathbf{A}(\theta_i) \mathbf{f}_r|^2 + \sigma^2 \right). \quad (8)$$

Because the variables to be optimized are coupled, we apply the alternating optimization (AO) algorithm to solve them.

A. Auxiliary Variable Optimization

For fixed $\{\mathbf{w}, \mathbf{f}_k, \mathbf{f}_r, \Phi\}$, it can be easily proven that $D(\mathbf{w}, \mathbf{f}_k, \mathbf{f}_r, \Phi, a)$ is concave with respect to a . Hence, by setting $\frac{\partial D}{\partial a} = 0$, we can obtain the optimal solution for a , given by

$$a^* = \frac{\eta_0 \mathbf{w}^H \mathbf{A}(\theta_0) \mathbf{f}_r}{\sum_{i=1}^L |\eta_i \mathbf{w}^H \mathbf{A}(\theta_i) \mathbf{f}_r|^2 + \sigma^2}. \quad (9)$$

B. Receive Beamforming Optimization

For fixed $\{\mathbf{f}_k, \mathbf{f}_r, \Phi, a\}$ and omitting the constant term in (8), the optimization problem can be rewritten as

$$(P2): \max_{\mathbf{w}} D_1(\mathbf{w}) \quad \text{s.t. } \|\mathbf{w}\|^2 = 1, \quad (10)$$

where

$$D_1(\mathbf{w}) = 2\Re \{ a^* \eta_0 \mathbf{w}^H \mathbf{A}(\theta_0) \mathbf{f}_r \} - |a|^2 \sum_{i=1}^L |\eta_i \mathbf{w}^H \mathbf{A}(\theta_i) \mathbf{f}_r|^2. \quad (11)$$

Let introduce the Lagrange multiplier μ for the constraint (10), the corresponding Lagrangian function is represented as

$$\mathcal{L}(\mathbf{w}, \mu) = D_1(\mathbf{w}) + \mu(1 - \|\mathbf{w}\|^2). \quad (12)$$

Then, we apply the Karush-Kuhn-Tucker (KKT) conditions [11] as follows,

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2\mathbf{B} - 2\mathbf{C}(\mu^*) \mathbf{w}^* = \mathbf{0}_N, \quad (13)$$

$$\mu^* (1 - \|\mathbf{w}^*\|^2) = 0, \quad (14)$$

where

$$\mathbf{B} = a^* \eta_0 \mathbf{A}(\theta_0) \mathbf{f}_r, \quad (15)$$

$$\mathbf{C}(\mu) = |a|^2 \sum_{i=1}^L |\eta_i|^2 \mathbf{A}(\theta_i) \mathbf{f}_r \mathbf{f}_r^H \mathbf{A}(\theta_i)^H + \mu \mathbf{I}_N. \quad (16)$$

Based on (13), the optimal solution for $\mathbf{w}$ is expressed as

$$\mathbf{w}^* = \mathbf{C}^{-1}(\mu^*) \mathbf{B}, \quad (17)$$

The optimal solution μ^* can be obtained from the following equation using the bisection search method:

$$\mathbf{B}^H (\mathbf{C}^{-1}(\mu^*))^H \mathbf{C}^{-1}(\mu^*) \mathbf{B} = 1. \quad (18)$$

C. Communication and Sensing Transmit Beamforming Optimization

For fixed $\{\mathbf{w}, \Phi, a\}$, problem (P1) can be reformulated as

$$(P3): \max_{\mathbf{f}_k, \mathbf{f}_r} D(\mathbf{f}_r)$$

$$\text{s.t. } (6a-6c),$$

$$|\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_k|^2 \geq \gamma_{th,k} \left(\sum_{j \neq k} |\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_j|^2 + |\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_r|^2 + \sigma_a^2 \|\mathbf{g}_k^H \Phi\|^2 + \sigma_u^2 \right), \forall k. \quad (19)$$

We first use a change of variables to simplify the problem. Define $\mathbf{F}_k = \mathbf{f}_k \mathbf{f}_k^H$, which satisfies $\mathbf{F}_k \succeq 0$ and $\text{rank}(\mathbf{F}_k) = 1, \forall k$. Since the rank-one constraint is non-convex, we use the SDR method to relax this constraint. As a result, problem (P3) can be reformulated as

$$(P4): \max_{\mathbf{F}_k, \mathbf{f}_r} D_2(\mathbf{f}_r)$$

$$\text{s.t. } \mathbf{F}_k \succeq 0, \quad (20a)$$

$$\sum_{k=1}^K \text{tr}(\mathbf{F}_k) + \mathbf{f}_r^H \mathbf{f}_r \leq P_{BS}, \quad (20b)$$

$$\sum_{k=1}^K \text{tr}(\mathbf{S} \mathbf{F}_k) + \mathbf{f}_r^H \mathbf{S} \mathbf{f}_r + \sigma_a^2 \|\Phi\|_F^2 \leq P_{ARIS}, \quad (20c)$$

$$\text{tr}(\mathbf{Q}_k \mathbf{F}_k) \geq \gamma_{th,k} \left(\sum_{j \neq k} \text{tr}(\mathbf{Q}_k \mathbf{F}_j) + \mathbf{f}_r^H \mathbf{Q}_k \mathbf{f}_r + \sigma_a^2 \|\mathbf{g}_k^H \Phi\|^2 + \sigma_u^2 \right), \forall k, \quad (20d)$$

where

$$D_2(\mathbf{f}_r) = 2\Re \{ a^* \eta_0 \mathbf{w}^H \mathbf{A}(\theta_0) \mathbf{f}_r \} - \mathbf{f}_r^H \mathbf{T} \mathbf{f}_r - |a|^2 \sigma^2, \quad (21)$$

$$\mathbf{T} = |a|^2 \sum_{i=1}^L |\eta_i|^2 \mathbf{A}^H(\theta_i) \mathbf{w} \mathbf{w}^H \mathbf{A}(\theta_i), \quad (22)$$

$$\mathbf{S} = \mathbf{G}^H \Phi^H \Phi \mathbf{G}, \quad (23)$$

$$\mathbf{Q}_k = (\mathbf{g}_k^H \Phi \mathbf{G})^H (\mathbf{g}_k^H \Phi \mathbf{G}). \quad (24)$$

As problem (P4) is a convex semidefinite program (SDP) over $\mathbf{F}_k$, it can be optimally solved by CVX [12].

D. Reflection Coefficients Optimization

For fixed $\{a, \mathbf{w}, \mathbf{f}_k, \mathbf{f}_r\}$, we define $\mathbf{V} = \mathbf{v} \mathbf{v}^H$, satisfying $\mathbf{V} \succeq 0$ and $\text{rank}(\mathbf{V}) = 1$. Note that by the change of variable [13], $\Phi \mathbf{G} \mathbf{f}_k = \mathbf{v}^H \text{diag}(\mathbf{G} \mathbf{f}_k)$, we have

$$\begin{aligned} \|\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_k\|^2 &= \text{diag}(\mathbf{f}_k^H \mathbf{G}^H) \mathbf{v} \mathbf{g}_k \mathbf{g}_k^H \mathbf{v}^H \text{diag}(\mathbf{G} \mathbf{f}_k) \\ &= \text{tr}(\mathbf{E}_{c,k} \mathbf{V}), \end{aligned} \quad (25)$$

where $\mathbf{E}_{c,k} = \text{diag}(\mathbf{f}_k^H \mathbf{G}^H) \mathbf{g}_k \mathbf{g}_k^H \text{diag}(\mathbf{G} \mathbf{f}_k)$. Similarly, we can obtain

$$\sum_{k=1}^K \|\Phi \mathbf{G} \mathbf{f}_k\|^2 + \|\Phi \mathbf{G} \mathbf{f}_r\|^2 + \sigma_a^2 \|\Phi\|_F^2 = \text{tr}(\mathbf{X} \mathbf{V}), \quad (26)$$

$$|\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_k|^2 = \text{tr}(\mathbf{E}_{c,k} \mathbf{V}), \quad (27)$$

$$\sum_{j \neq k} |\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_j|^2 = \text{tr}(\mathbf{U}_k \mathbf{V}), \quad (28)$$

$$|\mathbf{g}_k^H \Phi \mathbf{G} \mathbf{f}_r|^2 = \text{tr}(\mathbf{E}_{r,k} \mathbf{V}), \quad (29)$$

$$\sigma_a^2 \|\mathbf{g}_k^H \Phi\|^2 = \text{tr}(\tilde{\mathbf{G}}_k \mathbf{V}), \quad (30)$$

where

$$\begin{aligned} \mathbf{X} = & \sum_{k=1}^K \text{diag}(\mathbf{G} \mathbf{f}_k) \text{diag}(\mathbf{f}_k^H \mathbf{G}^H) \\ & + \text{diag}(\mathbf{G} \mathbf{f}_r) \text{diag}(\mathbf{f}_r^H \mathbf{G}^H) + \sigma_a^2 \mathbf{I}_M, \end{aligned} \quad (31)$$

$$\mathbf{E}_{c,k} = \text{diag}(\mathbf{f}_k^H \mathbf{G}^H) \mathbf{g}_k \mathbf{g}_k^H \text{diag}(\mathbf{G} \mathbf{f}_k), \quad (32)$$

$$\mathbf{U}_k = \sum_{j \neq k} \text{diag}(\mathbf{f}_j^H \mathbf{G}^H) \mathbf{g}_k \mathbf{g}_k^H \text{diag}(\mathbf{G} \mathbf{f}_j), \quad (33)$$

$$\mathbf{E}_{r,k} = \text{diag}(\mathbf{f}_r^H \mathbf{G}^H) \mathbf{g}_k \mathbf{g}_k^H \text{diag}(\mathbf{G} \mathbf{f}_r), \quad (34)$$

$$\tilde{\mathbf{G}}_k = \sigma_a^2 \text{diag}(\mathbf{g}_k^H) \text{diag}(\mathbf{g}_k). \quad (35)$$

Then, we apply the semidefinite relaxation (SDR) method to approximate the optimal solution of $\mathbf{v}$. Specifically, we can recover the corresponding rank-one solution by singular value decomposition (SVD) and take the eigenvector corresponding to the maximum eigenvalue. The problem can be rewritten as

$$\begin{aligned} \text{(P5):} \quad & \text{find } \mathbf{V} \\ \text{s.t.} \quad & \mathbf{V} \succeq 0, \end{aligned} \quad (36a)$$

$$\text{tr}(\mathbf{X} \mathbf{V}) \leq P_{\text{ARIS}}, \quad (36b)$$

$$\begin{aligned} \text{tr}(\mathbf{E}_{c,k} \mathbf{V}) \geq & \gamma_{th,k} \left[\text{tr}(\mathbf{U}_k \mathbf{V}) \right. \\ & \left. + \text{tr}(\mathbf{E}_{r,k} \mathbf{V}) + \text{tr}(\tilde{\mathbf{G}}_k \mathbf{V}) + \sigma_u^2 \right], \forall k, \end{aligned} \quad (36c)$$

$$\mathbf{V}_{m,m} \leq \alpha_{\max}^2, \quad m = 1, \dots, M. \quad (36d)$$

The overall algorithm is summarized as Algorithm 1.

IV. NUMERICAL RESULTS

In this section, simulation results verify the effectiveness of the algorithm and illustrate the advantages of the proposed active RIS-assisted ISAC system. Assume that the BS is located at the position (0,0). The RIS with $M = 8$ reflecting elements and $\alpha_{\max} = 20$ dB is located at (18 m, 3 m). Two users are randomly distributed within a circle centered at (30 m, 0) with a radius of 3 m. For simplicity, we assume a uniform communication SINR threshold for all users, i.e., $\gamma_{th,k} = \gamma_{th}, \forall k$. The BS-RIS and RIS-UE links are modeled as Rician channels with a Rician factor of 10 and Rayleigh fading channels [14], respectively. We consider communication in a

Algorithm 1 Proposed AO Algorithm to Solve Problem (P0).

Inputs: $\mathbf{G}, \mathbf{g}_k, \gamma_{th,k}, \forall k, P_{\text{BS}}, P_{\text{ARIS}}, \sigma^2, \sigma_a^2, \sigma_u^2, \alpha_{\max}, \eta_0, \eta_i, \forall i$.

Outputs: $\mathbf{w}^*, \mathbf{f}_k^*, \forall k, \mathbf{f}_r^*$, and Φ^* .

- 1: Initialize $\mathbf{w}, \mathbf{f}_k, \forall k, \mathbf{f}_r$ and Φ in a feasible region.
- 2: Set termination threshold ϵ and iteration index $l = 1$.
- 3: **repeat**
- 4: Update $a^{(l)}$ according to (9).
- 5: Update $\mathbf{w}^{(l)}$ according to (17) and (18).
- 6: Update $\mathbf{f}_r^{(l)}, \mathbf{f}_k^{(l)}, \forall k$, according to (P4).
- 7: Update $\Phi^{(l)}$ according to (P5).
- 8: Calculate $\text{SINR}_{s,opt}^{(l)} = \text{SINR}_s(a^{(l)}, \mathbf{w}^{(l)}, \mathbf{f}_r^{(l)}, \mathbf{f}_k^{(l)}, \Phi^{(l)})$.
- 9: $l \leftarrow l + 1$.
- 10: **until** $|\text{SINR}_{s,opt}^{(l)} - \text{SINR}_{s,opt}^{(l-1)}| / \text{SINR}_{s,opt}^{(l)} \leq \epsilon$ or the maximum number of iterations is reached.
- 11: **return** $\mathbf{w}^* = \mathbf{w}^{(l)}, \mathbf{f}_k^* = \mathbf{f}_k^{(l)}, \mathbf{f}_r^* = \mathbf{f}_r^{(l)}$, and $\Phi^* = \Phi^{(l)}$.

typical urban micro-cellular scenario, where the large-scale path loss is modeled using the ABG model, expressed as [15]

$$\text{PL}[\text{dB}] = 10\alpha \log_{10}\left(\frac{d}{1m}\right) + \beta + 10\gamma \log_{10}\left(\frac{f}{1\text{GHz}}\right) + \chi_{\sigma}^{\text{ABG}}. \quad (37)$$

The complex amplitude coefficient is modeled as $\eta = \sqrt{\frac{\lambda^2 S}{(4\pi)^3 R^4}}$, where $S = 1\text{m}^2$ [5]. The target of interest at an angle of 10° is at a distance of 30m from the BS. There are $L = 3$ interference sources at angles of $\theta_i \in \{-70^\circ, 40^\circ, 70^\circ\}$, each at a distance of 10m from the BS. The system total power budget is defined as $P_{\text{total}} = P_{\text{BS}} + P_{\text{ARIS}} + M(P_{\text{SW}} + P_{\text{DC}})$, where $P_{\text{SW}} = -5$ dBm and $P_{\text{DC}} = -10$ dBm. In the passive RIS-assisted ISAC system, the power budget is given by $P_{\text{total}} = P_{\text{BS}}^{\text{PRIS}} + M P_{\text{DC}}$ [5]. The carrier frequency of the ISAC system is 3 GHz.

The overall beampattern is defined as follows [16]:

$$P(\theta) = |\mathbf{w}^H \mathbf{A}(\theta) \mathbf{f}_r|^2. \quad (38)$$

In addition to the proposed deployment of active RIS, we include passive RIS with passive reflecting coefficients and random RIS with randomly adjusted coefficients for comparison. Fig. 2 illustrates the overall sensing beampattern. When $P_{\text{total}} = 15\text{W}$ and $\gamma_{th} = 5\text{dB}$ [2], we observe that, under the same system power budget, deploying active RIS in the ISAC system provides stronger gains in the direction of the target of interest while exhibiting comparable interference suppression capability.

We show the relationship between the sensing SINR and the communication SINR threshold in Fig. 3, which indicates that the active RIS scheme has better performance. As γ_{th} increases, the advantage of active RIS becomes more evident. Moreover, due to the increase in the total system power budget leading to an increase in the transmit power for sensing, a significant performance improvement is achieved for $P_{\text{total}} = 15\text{W}$ compared to $P_{\text{total}} = 10\text{W}$.

The sensing SINR versus the system power budget P_{total} is shown in Fig. 4. It can be observed that deploying the active RIS scheme has a significant advantage in sensing performance with the same system power budget. In addition, increasing the number of BS antennas N can bring excellent performance gains.

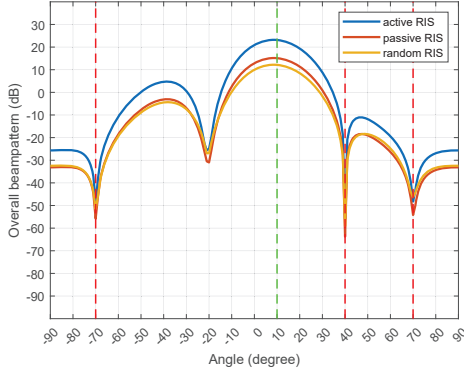


Fig. 2. Overall beampattern ($P_{\text{total}} = 15\text{W}$, $\gamma_{th} = 5\text{dB}$, $N = 4$).

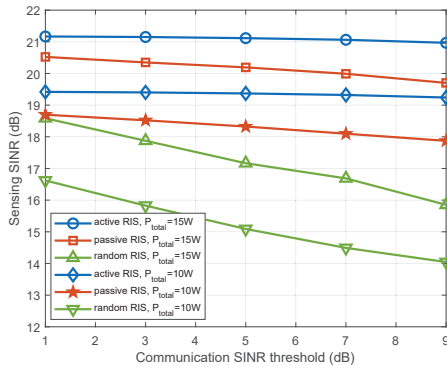


Fig. 3. Sensing SINR versus communication SINR threshold γ_{th} ($N = 4$).

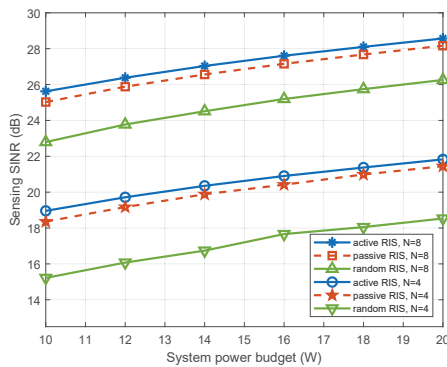


Fig. 4. Sensing SINR versus system power budget P_{total} ($\gamma_{th} = 5\text{dB}$).

V. CONCLUSION

This paper studies an active RIS-assisted ISAC system with sensing interference sources. The objective is to maximize

the sensing SINR while satisfying communication SINR constraints, the transmit power constraints of BS and active RIS, and unit-modulus constraints for receive beamforming. An algorithm based on QT, KKT, and SDR methods is proposed to transform the non-convex problem into several subproblems, which are then iteratively solved. Numerical results show that under the same power budget, deploying active RIS is a better strategy than both passive and random RIS schemes. In future work, we will try to use deep reinforcement learning to get better solutions.

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